

Project 05: Upcycle Challenge

Unit: Unit 5, Make It, Mend It, Wear It (Apparel and Textile Production and Design) **Topic:** 5.4 Upcycle Challenge project **Runs across:** Lessons 5.15 (the design brief), 5.16 (plan and prototype), 5.17 (make, two days), 5.18 (gallery and reflection, two days). Six days, days 136 to 141. **Team:** individual work, at a station of about seven with a station partner for help and peer review **Scored with:** Rubric 05 - Upcycle Challenge.md (five criteria, 20 points; the labs and projects category)

Standards this project assesses

- NYS ATPD 2. Production d) Develop sustainability practices for repurposing apparel items
- NYS ATPD 2. Production c) Demonstrate basic hand and machine sewing techniques by creating or repairing a textile product
- NYS ATPD 2. Production b) Demonstrate safe use and storage of equipment used in hand and machine sewing
- NYS ATPD 2. Production a) Identify common equipment and supplies used in hand and machine sewing
- NYS ATPD 4. Consumer Resources and Finance a) Compare and contrast the cost of constructing an apparel item versus purchasing one or repurposing one
- NYS ATPD 1. Selection c) Describe ways personal image and style is projected to others through clothing choices
- CTE PSI 2. Design Process (Proactive) a) Implement a formal design process to solve a given problem by a. Defining the problem being addressed; b. Defining criteria that must be met through the finished design; c. Defining constraints that must be adhered to; d. Brainstorming and examining possible solutions; e. Selecting the best solution for evaluation; f. Developing and constructing a prototype or model of the selected design; g. Testing and evaluating the prototype and model against the design criteria and constraints; h. Optimizing the solution for best form and function; i. Evaluating their use of the design process and how it impacted their final solutions
- CTE PSI 3. Troubleshooting Process (Reactive) a) Implement a formal troubleshooting process to solve a given problem by a. Defining the problem being addressed; c. Testing and evaluating to isolate the problem; d. Correcting the problem by implementing changes or repairs
- CTE SUS 1. Resources e) Investigate practices that promote stewardship of environmental resources
- CTE SUS 2. Environmental Management e) Explain factors to consider when assessing the environmental impact of purchasing decisions for everyday goods
- CTE HSW 4. Classroom and Workplace Safety c) Apply safety practices when using tools, machines, and equipment

- CTE HSW 4. Classroom and Workplace Safety e) Participate in housekeeping procedures that help to establish and maintain safe working conditions in the classroom or workplace
- CTE CIR 1. Communication f) Demonstrate personal development of communication skills through practice of these skills in a variety of classroom applications
- CTE CIR 2. Listening c) Show understanding of active listening and feedback techniques
- NYS FACS Standard 2 (Intermediate), Key Idea: Students will know the basic principles of home and community safety. They can demonstrate the skills necessary to maintain their homes and workplaces in a safe and comfortable condition. They can provide a safe and nurturing environment for themselves and others.; PI: Students use age-appropriate techniques to select and maintain clothing.
- CDOS Standard 2: Students will demonstrate how academic knowledge and skills are applied in the workplace and other settings.
- CDOS 3a Universal Foundation Skills: Thinking Skills; Personal Qualities; Interpersonal Skills; Managing Information
- National FCS 3.0: 16.4.5 Show the basic skills for making, altering, repairing, and recycling textiles and apparel.

The project sheet (student version)

Name: _____ Date: _____ Station: _____ Idea path: _____

The job

Find a used textile that was headed for the trash. Turn it into something a real person will actually use. Build it by hand in two class periods, using at least two of the four stitches you already know. Then explain it in 45 seconds to people who were not there while you made it.

You are not making a craft to take home and put in a drawer. You are making something with a user. The rubric asks who that person is, and “me” counts only if you can say what you will use it for.

The four constraints

Every project, every path, no exceptions.

1. **It must start from a used textile.** Something somebody already owned: a t-shirt, jeans, a towel, a pillowcase, a sweater, a sheet, fabric scraps. Not new fabric off a roll.
2. **It must be finished in two build days.** Lesson 5.17, two periods of 40 minutes. Your plan has to fit that, and your reduced-scope version is your insurance.
3. **It must use at least two of the taught joins:** running stitch, backstitch, whipstitch, sewing on a button, or a machine seam reverse-locked at both ends. Fabric glue and hem tape can hold one join if you need them, but two joins have to be stitched, by hand or on the machine.
4. **It must be something somebody would actually use, and you have to name who.** A name, and what they will use it for.

No sewing machine is needed anywhere in this project; every path can be built by hand. Since the machine block in Topic 5.2, the machines are also open on both build days under the same rules, and a machine seam counts as a taught join. Your plan sheet says which joins you are using, hand or machine, before anything is cut.

The ten example paths

Pick one, or design your own with my approval. The difficulty mark tells you how hard the construction is, not how good a grade it can earn. An Easy path built well scores higher than a Harder path built badly, every time.

#	Path	Starts from	Becomes	Stitches you will need	Difficulty
1	T-shirt tote bag, no sewing	A t-shirt	A bag for groceries, books, or gym clothes	None required for the body; whipstitch	Easy

				the handles to reinforce them and you meet constraint 3	
2	T-shirt headband	A t-shirt sleeve or body strip	A headband or hair tie	Whipstitch the ends into a loop, running stitch to gather	Easy
3	Sweater sleeve phone sleeve	One sleeve of a sweater or sweatshirt	A padded sleeve for a phone, glasses, or a calculator	Whipstitch the cut end closed, backstitch a reinforcement	Easy
4	Pillowcase drawstring bag	A pillowcase	A drawstring bag for shoes, laundry, or supplies	Running stitch for the drawstring channel, backstitch the side seam	Easy
5	Patchwork square	Fabric scraps, at least four	A mat, a potholder-style square, or a wall square	Backstitch to join the pieces, whipstitch the outer edge	Easy
6	Jeans-leg pencil case	One leg of a pair of jeans	A pencil case, tool roll, or brush case	Backstitch the bottom seam, whipstitch the edges, sew on a button for the flap	Medium
7	Shirt pillow cover	A button-front shirt	A pillow cover that uses the button placket as the opening	Backstitch three sides, whipstitch raw edges, button closure	Medium

				already there	
8	Towel produce bag	A hand towel or a cut piece of a bath towel	A washable bag for fruit and vegetables	Backstitch two sides, running stitch the top channel	Medium
9	Sock rice hand warmer	A clean sock with no holes	A hand warmer you heat at home	Running stitch to close, whipstitch the same seam a second time so nothing leaks	Medium
10	Sweatshirt cuff wristlet	The cuff of a sweatshirt	A wristlet or key pouch	Whipstitch one edge closed, backstitch a small inside pocket, button optional	Medium

Design your own. Bring me the idea with the four fields of the brief filled in and a sketch with measurements. If it fits the four constraints, it is approved. If it does not fit in two build days, I will tell you what to cut.

The rice hand warmer rules. Uncooked rice only. Nothing scented, nothing oily. It is never heated at school. Heating happens at home with an adult, 30 seconds maximum in a microwave, and your display card has to say so. If rice is a problem for you or anybody at home, use dried beans or flax seed instead.

What you turn in

Deliverable	What it is	When	Rubric criterion
1. The design brief	What you start with (what it is, its size in inches, the fiber if you can read the label, its condition), what it becomes, who will use it and for what, your two stitches, your tools,	Lesson 5.15	1. The plan and the design brief

	three steps in order, the one thing most likely to go wrong, and your reduced-scope version		
2. The approved plan sheet	A sketch with four or more measurements, the finished size and the cut size with the seam allowance shown, the three marks (cut, fold, stitch), the numbered order of work, the paper mock-up of the tricky part, what the mock-up showed and the one change you made, the peer review with one change you took and one you did not, and my initials	Lesson 5.16	1. The plan and the design brief
3. The finished item	Built to your plan, at least two of the taught joins (hand stitches or a machine seam), every seam closed, ends knotted and trimmed, closure working, no pins left in it, raw edges dealt with, and it does the job it was made for	Lesson 5.17 Day 2	2. Construction quality; 3. Useful and finished
4. Tool safety and the station routine	Count-out and count-in run correctly every day tools are out, nothing on the floor, nothing in a pocket, shears passed handle first, station	Lessons 5.16, 5.17 Day 1, 5.17 Day 2, 5.18 Day 1	4. Tool safety and station routine

	cleaned and materials bagged and labeled, all four project days		
5. The display card	Four fields in marker on card stock: WAS, NOW, STITCHES, HARD PART, plus EXTRA if you added a reinforcement or detail. Name on the back only.	Lesson 5.18 Day 1	5. The display card and explanation
6. The 45-second explanation	Four parts, pointing at the item: this was a ; now it is a for ; I used the stitch here and the ___ stitch here; the hard part was ___ and I fixed it by ___	Lesson 5.18 Day 2	5. The display card and explanation
7. The reflection	Five prompts answered: what it was and is; which two stitches and why those two; what went wrong and what you did; what you would change and keep with two more days; one thing you would tell a student about to do this project	Lesson 5.18 Day 2	1 and 5 (used as evidence)
8. The self-assessment row of Rubric 05	Your own score 1 to 4 on each of the five criteria with one line of evidence each, filled in before I score	Lesson 5.18 Day 2	Scored inside criterion 5

Also in the folder, not scored on the rubric: the paper mock-up, the peer review checklist you filled for your station partner, the two or more feedback slips you wrote for other

people, and the slips other people wrote for you. A missing mock-up or missing slips is noted on the rubric's notes line and counts in the daily work category.

Day by day

Day	Lesson	What happens	Checkpoint
1 (day 136)	5.15	Upcycle against recycle. The ten paths. The four constraints. Pick a material from the scrap bin, the donation drive, or home with a signed note. Write the design brief, including your reduced-scope version. Rubric goes out.	Brief names a used material with a measurement, two of the taught joins, and a named user
2 (day 137)	5.16	Plan sheet with measurements, finished size and cut size, seam allowance, the three marks, the order of work. Paper mock-up of the tricky part. Approval at my table. Peer review. Nothing is cut before I initial the plan.	My initials on the plan sheet; the mock-up built and the one change written
3 (day 138)	5.17 Day 1	Count-out. Cut on the marked line, mark the stitch line, pin or clip, stitch the first full seam or edge. Stuck protocol. Milestone check. Count-in.	Cut, marked, pinned, one full seam or edge stitched
4 (day 139)	5.17 Day 2	Count-out. Finish the item. Reduced-scope decision by minute 10 if you are	All seven finishing lines pass, or the remaining work is written down and

		behind. Seven-line finishing checklist, self-checked and initialed. Count-in.	scheduled
5 (day 140)	5.18 Day 1	Ten-minute finishing window for anyone who needs it. Display card in marker. The 45-second planner. Gallery set up on three tables. Practice the 45 seconds twice, timed, with a partner.	Display card complete on all four fields; the 45 seconds fits in 45 seconds
6 (day 141)	5.18 Day 2	Gallery walk with at least two feedback slips. The 45-second explanations in rotation. The vote on three categories. The reflection. Self-assessment row. Folder and item turned in.	Folder complete; the explanation given at least three times, or once one-on-one

Day 143, the last day of the unit, returns the item and the rubric after the Unit 5 Test.

Where you get to choose

- **The path.** Ten examples plus design-your-own. Yours to pick. You may change paths up to the end of Lesson 5.16 with no penalty; after your plan is initialed and your fabric is cut, you switch to your reduced-scope version instead, which is also no penalty.
- **The material.** Anything in the scrap bin, anything the donation drive brings in, or something of your own with a signed note. Nothing in the bin is claimed until you pick it up.
- **The user.** Who the item is for is entirely yours: yourself with a stated use, a family member, a friend, a teacher, the classroom, a younger sibling, a pet.
- **The stitches.** Any two of the four, wherever they make sense. A third stitch is a stretch, not a requirement.
- **How you show it.** The 45 seconds can be given to a rotating audience, to one partner at your own table, or one-on-one to me. The display card can be handwritten, typed, or dictated to me word for word.

- **How much you add.** A reinforcement at a stress point, a second row of stitching, a hand-stitched detail, or a second small item if you finish early. Say what you added on the card and it counts on criterion 2.

The sourcing plan, and the no-item-from-home path

Three ways to get a material, and nobody is stuck:

1. **The room's scrap bin.** Free, open from day 116, and stocked with more pieces than there are students. Anything in it is yours. **This is the default and it is a full-credit path.** A student who brings nothing from home builds exactly the same project and is scored on exactly the same rubric. Nobody is asked in front of the class whether they brought something.
2. **The class donation drive.** The bin by the door takes clean, used textiles from anybody, and everything that comes in goes into the scrap bin for the whole class. Contributing is not required and is not tracked.
3. **An item from home, with the signed permission note.** The note is the last page of Handout 05.15. Take it home, have the owner sign it, bring both. No signed note, no cutting, and the bin is right there.

Nothing valuable, nothing borrowed, nothing that belongs to somebody who has not signed. Everything clean and dry. If you bring something and then decide you do not want to cut it, that is fine and you use the bin.

Supports

- **The picture plan template** (on the back of Handout 05.15). The brief and the plan can be done as a labeled drawing with measurements instead of sentences. Every field has a picture box and a number blank. This is a full-credit version of the plan, not a reduced one, and it is scored on the same criterion 1 descriptors.
- **Sentence starters.** The Turkish, Portuguese, and Spanish lines are to be verified by a native speaker.

English	Turkish	Portuguese	Spanish
This was a ____.	Bu bir ____ idi.	Isto era uma ____.	Esto era un ____.
Now it is a ____ for ____.	Şimdi ____ için bir ____.	Agora é uma ____ para ____.	Ahora es un ____ para ____.
I used the running stitch here.	Burada teyel dikişi kullandım.	Usei o ponto corrido aqui.	Usé la puntada corrida aquí.
I used the backstitch here because it is strong.	Burada geri dikiş kullandım çünkü sağlamdır.	Usei o ponto atrás aqui porque é forte.	Usé la puntada atrás aquí porque es fuerte.
The hard part was ____.	En zor kısım ____ oldu.	A parte difícil foi ____.	La parte difícil fue ____.
I fixed it by ____.	____ yaparak düzelttim.	Consertei ____.	Lo arreglé ____.

___ will use it to ___.	___ bunu ___ için kullanacak.	___ vai usar para ___.	___ lo va a usar para ___.
My item is finished.	Ürünüm bitti.	Meu item está pronto.	Mi artículo está terminado.

- **The reduced-scope version.** Written by you on your own brief on day 136, before you need it. If you are behind at minute 10 of Lesson 5.17 Day 2, you switch to it and you are scored on the same five criteria for what you actually made. The common reductions are printed on Handout 05.17.
- **Reduced-dexterity tools.** Self-threading needles, large-eye needles with heavier thread, needle threaders, pre-threaded needles ready at my table, binder clips instead of pins, a clamp or tape to hold the fabric, felt instead of fraying woven fabric, fabric glue or iron-on hem tape for one of the two joins, and a larger item with longer stitches instead of a small item with small ones. Using any of these does not change your score or appear on it.
- **Pre-measured and pre-marked material.** Available from me, cut from the same scrap bin, for any student whose plan calls for it.
- **The no-presentation option.** The written display card plus a 45-second explanation one-on-one to me, at your own station or the side table. Scored identically on criterion 5. Offered privately, not announced.
- **Extended time.** Any deliverable may finish in the first five to ten minutes of the next lesson. The ten-minute finishing window at the start of Lesson 5.18 Day 1 exists for the item. The whole folder is due at the end of Lesson 5.18 Day 2; a late project is accepted with a one-level drop on one criterion, never a zero, per the grading plan.
- **If you are absent** on a build day: the written alternative is the plan sheet completed in full plus a written step-by-step construction procedure for your item, and then you build during the Lesson 5.18 Day 1 window and any available time I can give you. Nobody loses points for being out.

How it is scored

Rubric 05 - Upcycle Challenge.md has five criteria, each scored Not Yet (1), Approaching (2), Meets (3), or Exceeds (4), for 20 points:

1. The plan and the design brief
2. Construction quality, including the stitches used
3. The item is genuinely useful and finished
4. Tool safety and the station routine, across all four days
5. The display card and the explanation

You fill the self-assessment row first, with one line of evidence per criterion, before I score. The rubric goes out on day 136 with this sheet, so you build against it.

The project counts in the labs and projects category (40 percent of the marking period). Criterion 4 also feeds the participation and safety category. The Unit 5 Test on day 143 includes a repair decision and a sustainability item, so keep your reflection in mind.



Teacher notes on the project

- This is the unit's synthesis project and it is deliberately smaller in scope than the Unit 4 room design. Unit 4 asked students to plan a space on paper; this one asks them to make a physical object that works, with their hands, in 80 minutes of build time. The whole design is built around that constraint: an approval gate before anyone cuts, a milestone at the end of each build day, a stuck protocol so one teacher can serve 21 makers, and a reduced-scope version written on day one so no student has to invent a fallback while panicking.
- The state's Sustainability theme module names this project almost exactly. Its Family and Consumer Sciences illustrative activity ends in "a recycle/upcycle/repurposing project, such as recycling t-shirts into reusable shopping bags," which is path 1 on the menu. The ATPD content module line 2. d), "Develop sustainability practices for repurposing apparel items," is the only place in the whole framework where the words repurposing apparel appear, and this project is the answer to it.
- The scrap bin is the single most important piece of preparation. Start it on day 116, the first day of the unit, with a sign. By day 136 it should hold more usable pieces than there are students. A bin with four pieces in it makes twenty students fight, and it makes the students with nothing at home feel it.
- Say the default out loud on day 136, before anyone asks: "Most people will use the bin. That is the normal way to do this project." That one sentence is why the no-item-from-home path works.
- The approval gate is not optional. Fabric shears live at the teacher's table until a plan is initialed. Five questions, sixty seconds, every student: used material, two taught stitches, cut size includes the seam allowance, order of work is possible, reduced scope written. That conversation prevents almost every failure this project can have.
- Scope is the enemy. The students who finish with nothing are the ones who planned a backpack, a quilt, or a stuffed animal. Constraint 2 and the reduced-scope line exist for them. Steer on day 136, not on day 139.
- Individual work with station partners is intentional. A group project on a sewing build produces one student sewing and three watching. Everyone makes their own item; the station is for help, tools, and peer review.
- Criterion 4 is scored from the clipboard across four days, not from one observation. Three needle and pin counts per day (one per station) are the evidence, and a student who runs a count correctly as Station Manager gets that noted. Rotate Station Manager daily so three different students run a count each day.
- Grade the item on what it is, not on how ambitious it was. A whipstitched phone sleeve with even stitches and a working fit scores higher than a half-built pencil case with a great idea. Say that on day 136 and students choose honestly.
- Photograph every item with its display card on day 141, no names and no faces. That photo set is the Unit 5 page on the website, the hallway display case, and next year's idea menu slides.
- **Material cost for the unit budget.** The textiles are free (scrap bin plus donation drive). The consumables this project adds beyond the Topic 5.2 sewing kit: cotton cord or shoelaces for drawstring bags (a spool, about \$4), assorted buttons (a jar, about \$6,



or free from the drive), uncooked rice (a two-pound bag, about \$2), fabric glue (one bottle, about \$4), iron-on hem tape (one roll, about \$4), binder clips (a box of 12, about \$2), card stock for display cards (one pack, about \$6, or the copier's heaviest paper for free), butcher paper or three cloths for the display tables (about \$8 or free). Total new spend for the project, about \$20 to \$36 for a class of 21, or under \$2 per student. Everything else (needles, pins, thread, shears, seam rippers, thimbles, threaders, chalk) is already in the Topic 5.2 line and is reused every year.